

CrowdSense

Real-time crowd risk detection from CCTV using YOLOv8 and density estimation

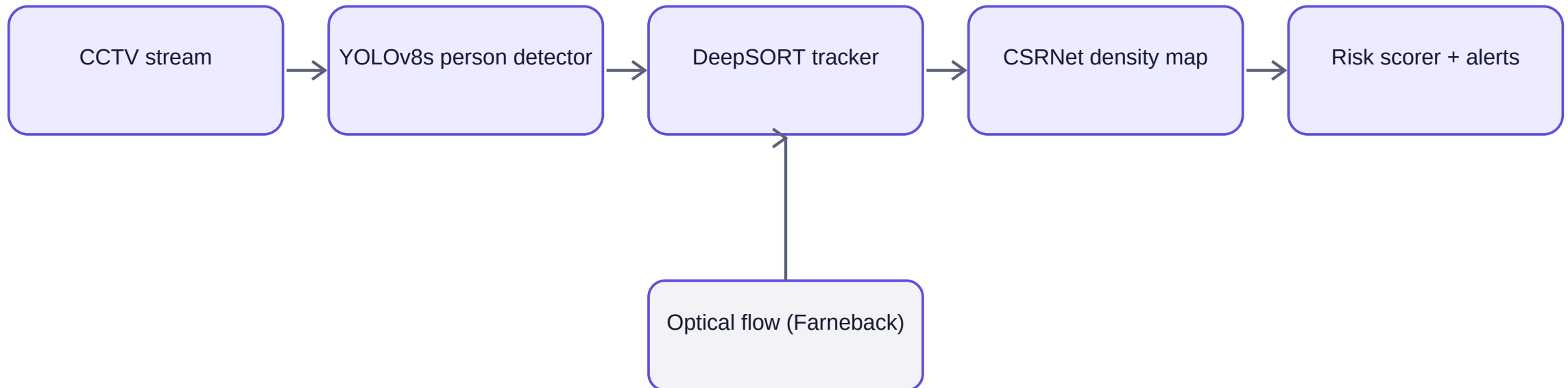
Final year project viva | Department of CSE

Problem

- Crowd crushes caused more than 1,400 deaths worldwide between 2010 and 2023
- Manual CCTV monitoring: one operator watches around 40 camera feeds
- Existing systems count people but do not predict dangerous density build up
- Goal: raise an alert at least 60 seconds before density crosses 5 people per square metre

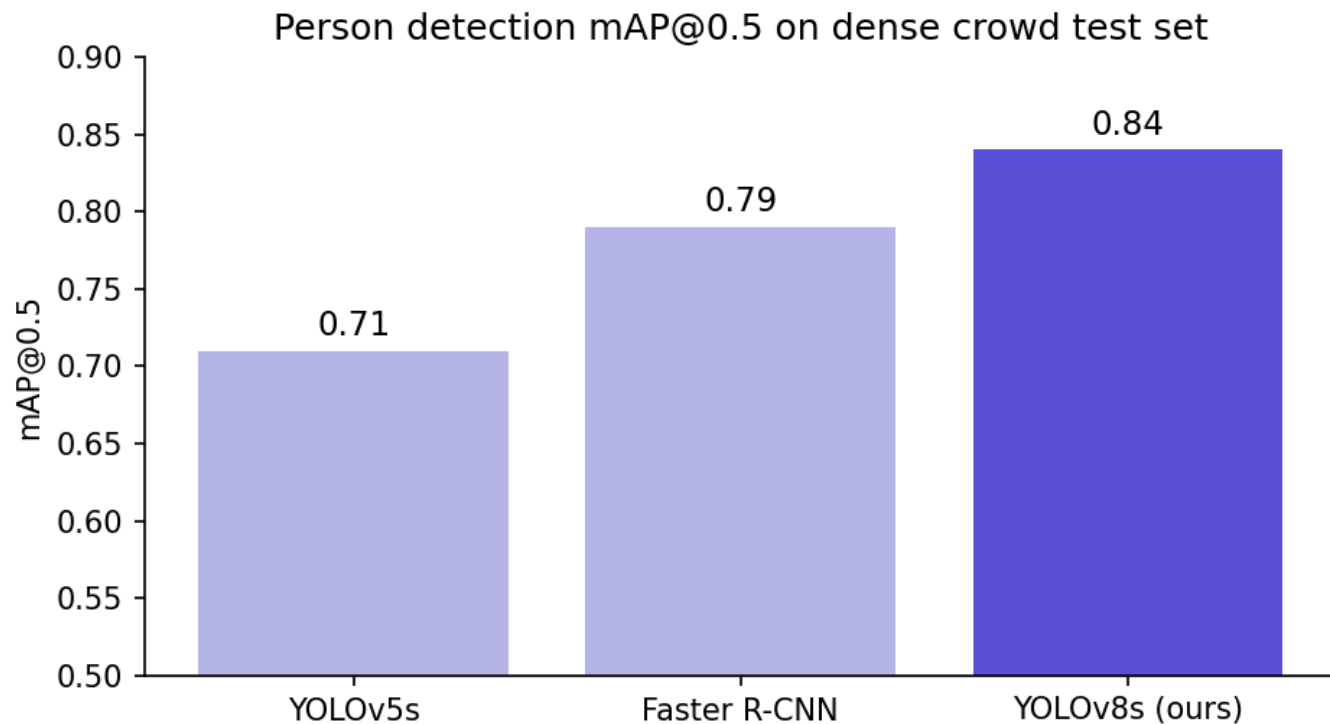
System Architecture

End to end pipeline running at 38 FPS on an RTX 3060



Risk score = density x flow turbulence, smoothed over a 10 s window

Detection Results



- YOLOv8s improves mAP by 13% over YOLOv5s
- Runs at 38 FPS vs 9 FPS for Faster R-CNN
- Trained on 4,200 annotated frames

Figure: detection accuracy comparison on our dense crowd test split

Density Estimation Evaluation

Mean Absolute Error (lower is better)

Dataset	MCNN	CSRNet (ours)	Improvement
ShanghaiTech A	110.2	68.2	38%
ShanghaiTech B	26.4	10.6	60%
UCF-QNRF	277.0	135.4	51%

Risk alerts: 92% precision, 87% recall on 36 annotated incident clips

Limitations and Future Work

- Night time footage reduces detection recall to 0.61
- Heavy occlusion above 8 people per square metre breaks tracking
- Future: thermal cameras, edge deployment on Jetson Orin
- Future: evaluate on live festival footage with the city police control room